Assignment: Building Dashboard & Storytelling

Q1. Difference between a Tableau Dashboard and a Tableau Story Point.

Answer

A Tableau Dashboard is a single interactive screen that combines two or more worksheets, KPI cards, legends and filters. It is mainly used for monitoring and comparing several measures at the same time. A Tableau Story is a sequence of worksheets or dashboards arranged as Story Points. It is used to explain findings step by step, similar to a presentation.

Feature	Dashboard	Storypoint
Purpose	Monitor several visuals together	Communicate an analytical sequence
Layout	One screen containing multiple sheets	Multiple points shown one after another
Interaction	Filters, highlights and actions	Navigation through a narrative
Healthcare u	Daily hospital performance dashboard	Patient journey or cost-analysis presentation

Healthcare use case 1: A dashboard can show total patients, department cost, outcome percentages and regional patient volume on one screen so hospital managers can monitor performance.

Healthcare use case 2: A story can begin with regional patient distribution, move to department cost, then explain outcomes and finally show doctor performance. This makes the analysis easier for management to follow.

Q2. Actions in Tableau and the Use of Filter Action.

Answer

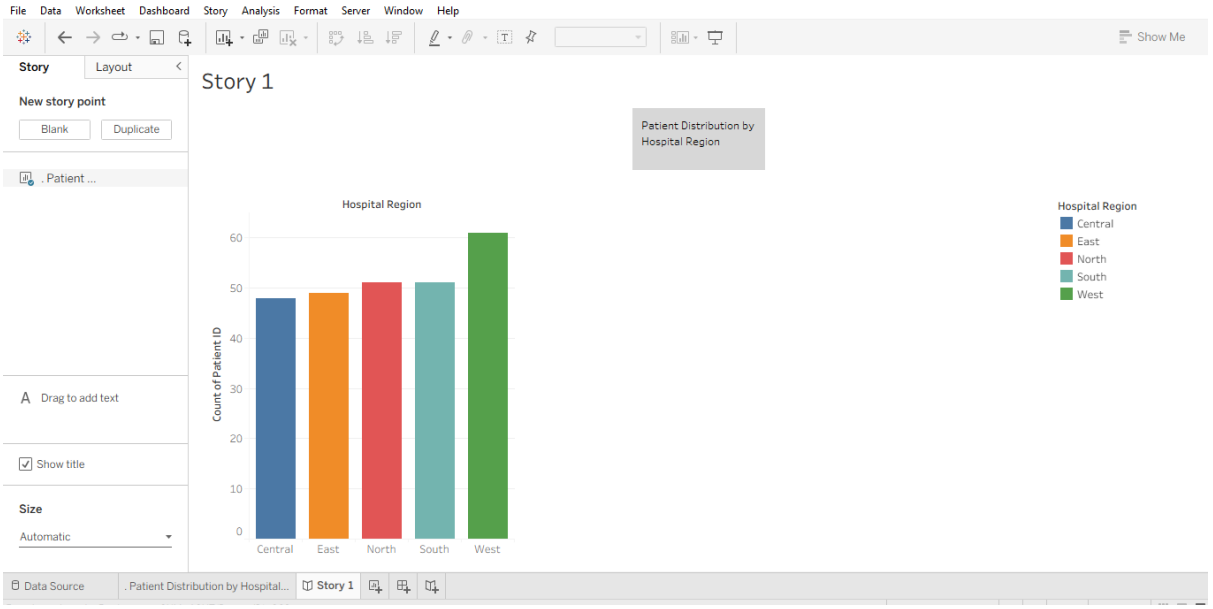
Actions make a Tableau dashboard interactive. The main actions are Filter Action, Highlight Action, URL action, Go to Sheet Action and Parameter Action.

A Filter Action uses a selected mark in one chart to filter data shown in other charts.

For example, when a manager clicks the West region in the Patient Distribution chart, the Department Cost and Outcome charts can automatically update to show only West-region records. This allows management to compare hospital regions without creating separate dashboards. The same approach can be used for Department, Doctor, Outcome or Treatment Type.

Q3. Patient Distribution by Hospital Region

Answer

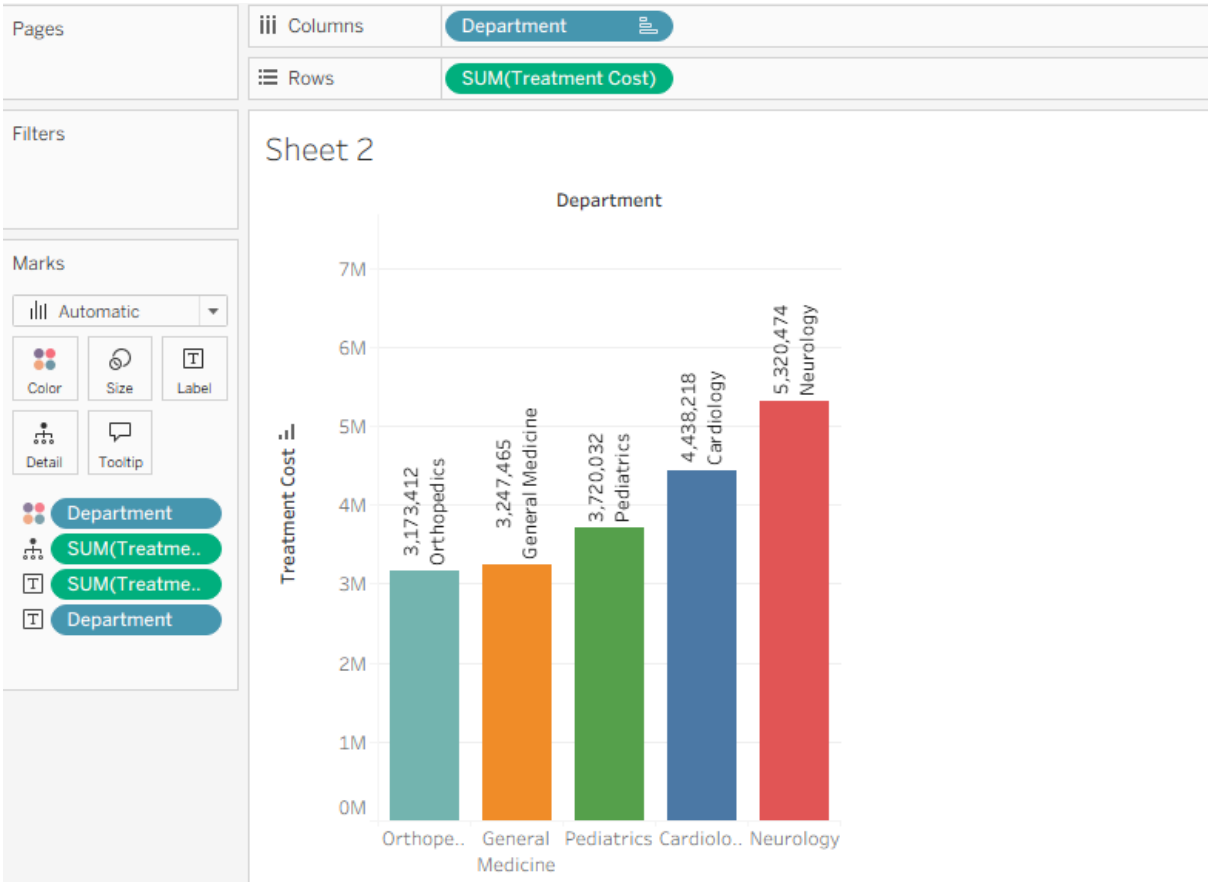


Insights:

- West has the highest patient volume with 61 patients.
- The difference between the largest and smallest regions is 13 patients.
- The regional pattern can help management allocate beds, staff and diagnostic resources according to demand.

Q4. Department Performance - Treatment Cost by Department

Answer

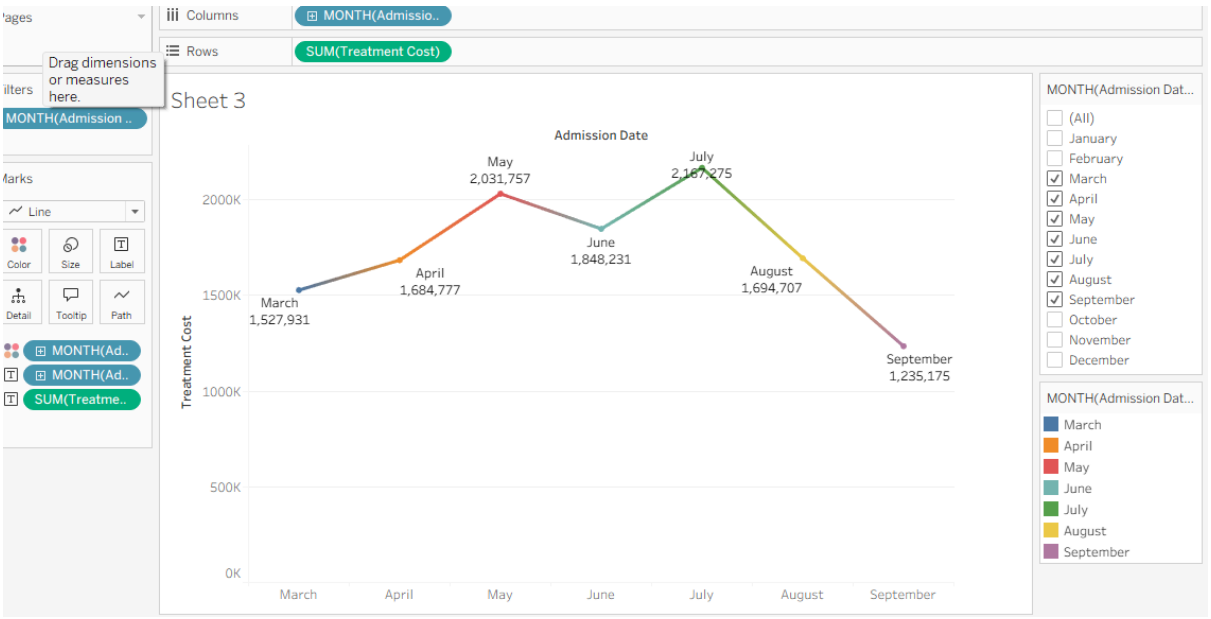


Insights:

- Neurology generated the highest total treatment cost of ₹5,320,473.65.
- The lowest-cost department was Orthopedics with ₹3,173,411.69.
- A high treatment cost may indicate greater case complexity, higher patient volume or more expensive procedures; management should compare this with outcomes and occupancy before making decisions.

Q5. Treatment Cost Trend Over Time - March to September.

Answer

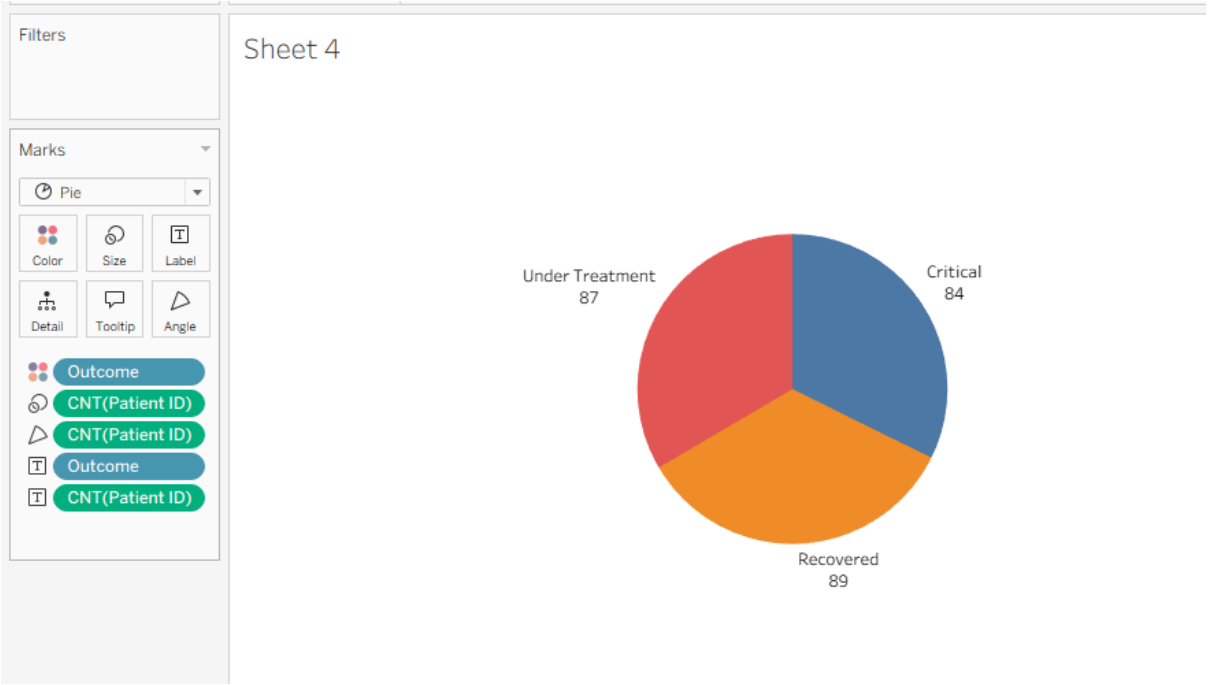


Insights:

- July recorded the highest treatment cost in the selected period at ₹2,167,275.49.
- September recorded the lowest treatment cost at ₹1,235,174.96.
- Monthly cost variation can support cash-flow planning, staffing schedules and the purchase of medicines and medical supplies.

Q6. Patient Outcome Analysis.

Answer



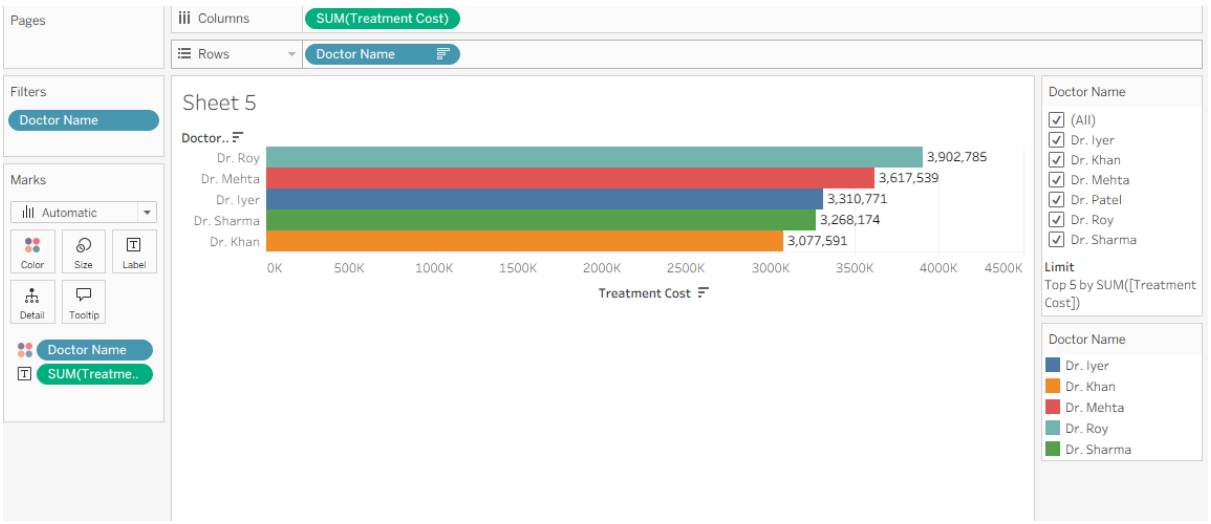
Insights:

- Recovered is the largest outcome category with 89 patients.
- There are 84 critical patients, which should be monitored closely by region and department.

Q7. Doctor Performance and Patient KPI Analysis

Answer

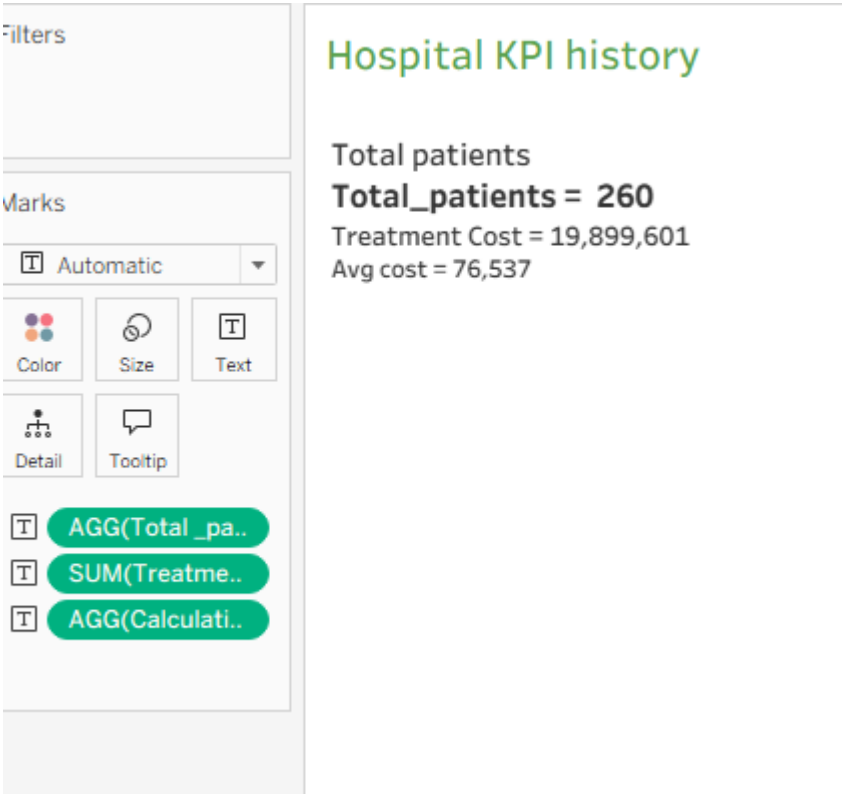
Part A - Doctor Performance



Insights:

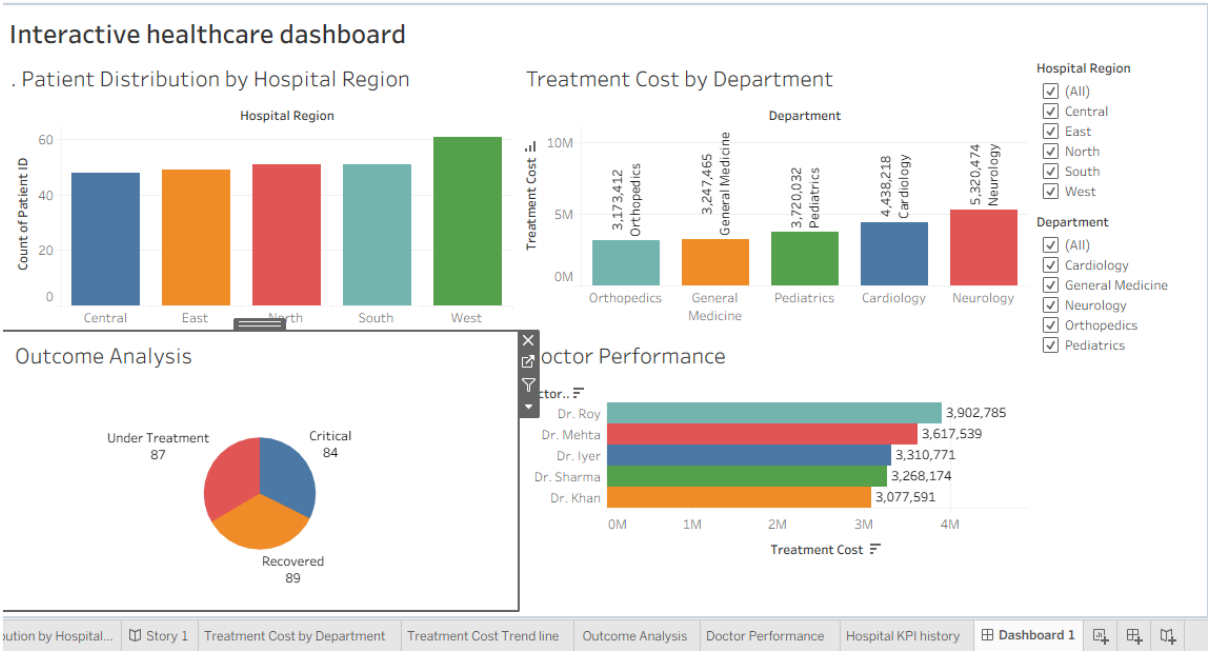
- Dr. Roy handled the highest total treatment cost at ₹3,902,785.36.
- Treatment cost should not be used as the only measure of doctor performance. It should be viewed together with patient count, outcomes, case complexity and length of stay.

Part B - Total Patients Treated KPI



Q8. Interactive Hospital Management Dashboard

Answer



Insights:

- The dashboard allows managers to move from a hospital-wide view to a selected region or department.
- KPI cards provide immediate context before the detailed charts are reviewed.
- The dashboard can be used in monthly operational reviews to discuss cost, patient load and quality outcomes.

Q9. Tableau Storytelling Flow

Answer

I created the following four-point story flow so that the audience first understands demand, then cost, then clinical outcomes and finally doctor-level performance.

Story Point 1 - Patient Distribution

Shows which hospital region treats the most patients and helps identify high-demand locations.

Story Point 2 - Department Cost Analysis

Compares total treatment cost by department and highlights the department creating the greatest financial workload.

Story Point 3 - Outcome Analysis

Explains the percentage and number of recovered, critical and under-treatment patients.

Story Point 4 - Doctor Performance

Ranks doctors by the total treatment cost handled and provides a management view of workload.

Each Story Point should have a clear caption explaining what the audience should observe. The final Story Point can use the dashboard as a summary of the full analysis.

Q10. Final Findings and Business Recommendations

Answer

1. Which department generates the highest treatment cost?
Neurology (₹53.2 lakhs total) — by a clear margin over Cardiology.

2. Which region has more critical patients?
West (tied at 20 critical patients each) — highest critical caseload.

3. What business decisions can hospital management take?

- Since Neurology costs the most, review pricing/insurance packages and ensure adequate specialist staffing and equipment there.
- Since West have the most critical patients, allocate more ICU beds, critical-care doctors, and emergency resources to these regions.

- Dr. Roy and Dr. Mehta handle the highest-cost caseloads — worth reviewing their case complexity/workload distribution to prevent burnout or bottlenecks.